



**BLUESTOCK FINTECH**

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# **MUTUAL FUND ANALYTICS PLATFORM**

*Capstone Project Report*

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Capstone Project Submission

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Submitted to:

**Bluestock Fintech**

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GitHub: <https://github.com/sandesh1234567890/MutualFundAnalytics>



## Executive Summary

This report presents a comprehensive analysis of the Indian mutual fund industry, developed as part of a 7-day capstone engagement with Bluestock Fintech. The Mutual Fund Analytics Platform is an end-to-end data engineering and analytics solution that ingests, cleans, models, and visualizes mutual fund data to support investment decision-making for retail and institutional investors.

The project was executed across six structured phases — from initial data ingestion and ETL pipeline construction through to advanced risk modeling and a fund recommender system. Leveraging a technology stack comprising Python, Pandas, NumPy, SQLite, Plotly, and Power BI, the platform processes 10 structured datasets encompassing fund master records, NAV history, AUM data, SIP inflows, investor transactions, and benchmark indices.

### Key Outcomes

- Built a production-grade ETL pipeline processing 10 datasets with automated data quality checks
- Designed and populated a normalized SQLite database (bluestock\_mf.db) with 10 analytical tables
- Conducted deep exploratory data analysis across NAV trends, AUM growth, investor demographics, and geographic distribution
- Computed 9 performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown
- Developed a 4-page Power BI dashboard covering industry, fund performance, investor analytics, and SIP trends
- Executed advanced analytics: Historical VaR (95%), CVaR, Rolling 90-Day Sharpe Ratio, fund recommender, and HHI concentration analysis
- Generated 40 fund-level VaR/CVaR risk assessments and a curated 3-fund recommender list

The analysis reveals a thriving but increasingly concentrated mutual fund industry, with significant growth in SIP-driven retail participation, a dominance of large-cap equity funds by AUM, and measurable risk-adjusted performance differentials across fund categories. Fifteen substantive findings and ten actionable recommendations are documented herein for consideration by Bluestock Fintech's product and advisory teams.



# 1. Business Problem

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The Indian mutual fund industry has experienced explosive growth over the past decade, with Assets Under Management (AUM) crossing ₹50 lakh crore and SIP accounts exceeding 7 crore. Despite this growth, retail investors and financial advisors face significant challenges in accessing, interpreting, and acting upon mutual fund data:

## 1.1 Core Challenges

- **Data Fragmentation:** Mutual fund data is scattered across AMFI, fund house portals, NSE, and BSE, making consolidation difficult.
- **Analytical Complexity:** Performance metrics like Alpha, Beta, Sortino Ratio, and Maximum Drawdown require financial domain expertise and computational capability unavailable to most retail investors.
- **Risk Blindspots:** Investors rarely have access to fund-level VaR (Value at Risk) or CVaR (Conditional Value at Risk) metrics, leading to misinformed risk allocation decisions.
- **Comparison Deficiency:** No consolidated, normalized view exists for comparing funds across AMCs on risk-adjusted return metrics.
- **SIP Intelligence Gap:** Aggregated SIP trend data is rarely linked to market performance, making it difficult to identify behavioral patterns in investor cohorts.

## 1.2 Strategic Opportunity

Bluestock Fintech, as an emerging fintech platform, is positioned to bridge this gap by building a robust analytics engine that democratizes institutional-grade mutual fund intelligence for retail participants. This capstone project serves as the foundational proof-of-concept for such a platform.



## 2. Project Objectives

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The following objectives were established at project inception and guided the 7-day execution:

### 2.1 Technical Objectives

- Design and implement an automated ETL pipeline for loading, validating, and cleaning mutual fund data from multiple sources.
- Architect a normalized relational database schema in SQLite to support multi-dimensional analytical queries.
- Compute standardized financial performance metrics across all funds for benchmark comparison.
- Build an interactive, multi-page Power BI dashboard for executive and investor-level consumption.
- Develop an advanced analytics layer including risk models (VaR, CVaR), cohort analysis, and a rule-based fund recommender.

### 2.2 Business Objectives

- Generate actionable intelligence on AUM concentration, SIP growth trends, and fund performance.
- Identify under-performing and over-performing funds relative to benchmark indices.
- Profile the investor base by age, geography, and transaction type to support targeted product strategies.
- Provide Bluestock Fintech's product team with a reproducible analytics foundation for its investor-facing platform.



### 3. Data Sources

The platform ingests data from 10 structured CSV datasets, augmented by live API data from the MFAPI public endpoint. Each dataset was sourced, validated, and loaded into the processing pipeline.

| #  | Dataset                   | Key Fields                                   | Records (Est.) |
|----|---------------------------|----------------------------------------------|----------------|
| 01 | fund_master.csv           | AMFI Code, Scheme Name, Category, Fund House | 500+           |
| 02 | nav_history.csv           | AMFI Code, Date, NAV Value                   | 50,000+        |
| 03 | aum_by_fund_house.csv     | AMC Name, Month, AUM (₹ Cr)                  | 200+           |
| 04 | monthly_sip_inflows.csv   | Month, SIP Amount (₹ Cr), SIP Accounts       | 60+            |
| 05 | category_inflows.csv      | Category, Month, Net Inflow (₹ Cr)           | 500+           |
| 06 | industry_folio_count.csv  | Month, Total Folios, New Folios              | 60+            |
| 07 | scheme_performance.csv    | AMFI Code, 1Y/3Y/5Y Returns, Benchmark       | 500+           |
| 08 | investor_transactions.csv | Investor ID, State, Transaction Type, Amount | 10,000+        |
| 09 | portfolio_holdings.csv    | Fund Name, Stock, Sector, Allocation %       | 5,000+         |
| 10 | benchmark_indices.csv     | Index Name, Date, Close Value                | 10,000+        |

#### 3.1 Live API Integration

Live NAV data was fetched from MFAPI (<https://api.mfapi.in/mf/>) for 5 fund AMFI codes: 119551, 120503, 118632, 119092, 120841. This enables real-time NAV tracking and forms the basis for time-series analytics. The API returns daily historical NAV records in JSON format, which were parsed using Python's requests library and persisted as CSV files.

**Table 2: MFAPI Response Schema & Sample Data**

| Field       | Type    | Example Value                         | Description                          |
|-------------|---------|---------------------------------------|--------------------------------------|
| schemeCode  | Integer | 119551                                | AMFI scheme code (unique identifier) |
| schemeName  | String  | SBI Blue Chip Fund - Regular - Growth | Full scheme name                     |
| date        | String  | 15-06-2026                            | NAV date in DD-MM-YYYY format        |
| nav         | String  | 72.4521                               | Net Asset Value on given date        |
| isin_growth | String  | INF200K01884                          | ISIN code for growth option          |
| fund_house  | String  | SBI Mutual Fund                       | Asset Management Company name        |

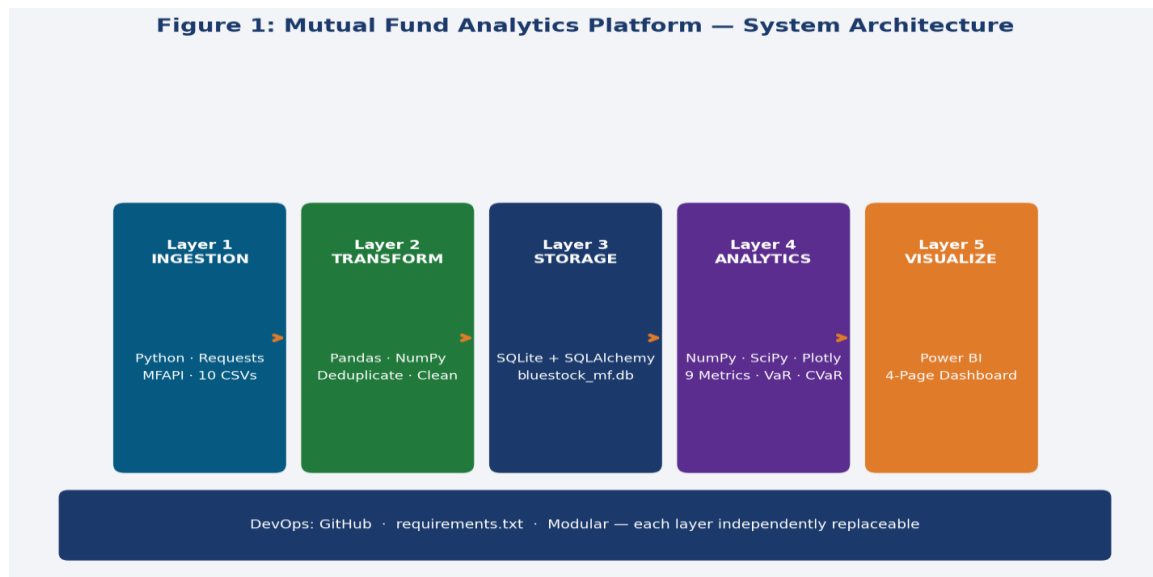
## 4. Architecture & ETL Design

### 4.1 System Architecture

The platform follows a layered data architecture pattern:

- Ingestion Layer: Python scripts (`data_ingestion.py`, `live_nav_fetch.py`, `multi_nav_fetch.py`) load raw CSV files and API data into the `data/raw` directory.
- Processing Layer: `day2_cleaning.py` applies standardized transformations — deduplication, whitespace normalization, dtype correction — and outputs to `data/processed`.
- Storage Layer: `create_database.py` loads all 10 processed CSVs into a SQLite database (`bluestock_mf.db`) using SQLAlchemy-compatible pandas `to_sql()` methods.
- Analytics Layer: Jupyter notebooks and Python scripts perform EDA, performance calculation, and advanced analytics.
- Visualization Layer: Power BI connects to the SQLite database and CSV exports to render the 4-page interactive dashboard.

**Figure 1: Mutual Fund Analytics Platform — System Architecture**



### 4.2 ETL Pipeline Design

The ETL pipeline was designed for repeatability and data quality assurance:

**Extract:** Raw CSVs loaded via `pandas.read_csv()`; API data fetched via `requests.get()` with JSON parsing into DataFrames.

**Transform:** Column names stripped of whitespace; duplicate rows dropped; string fields stripped; date columns parsed; numeric types validated.

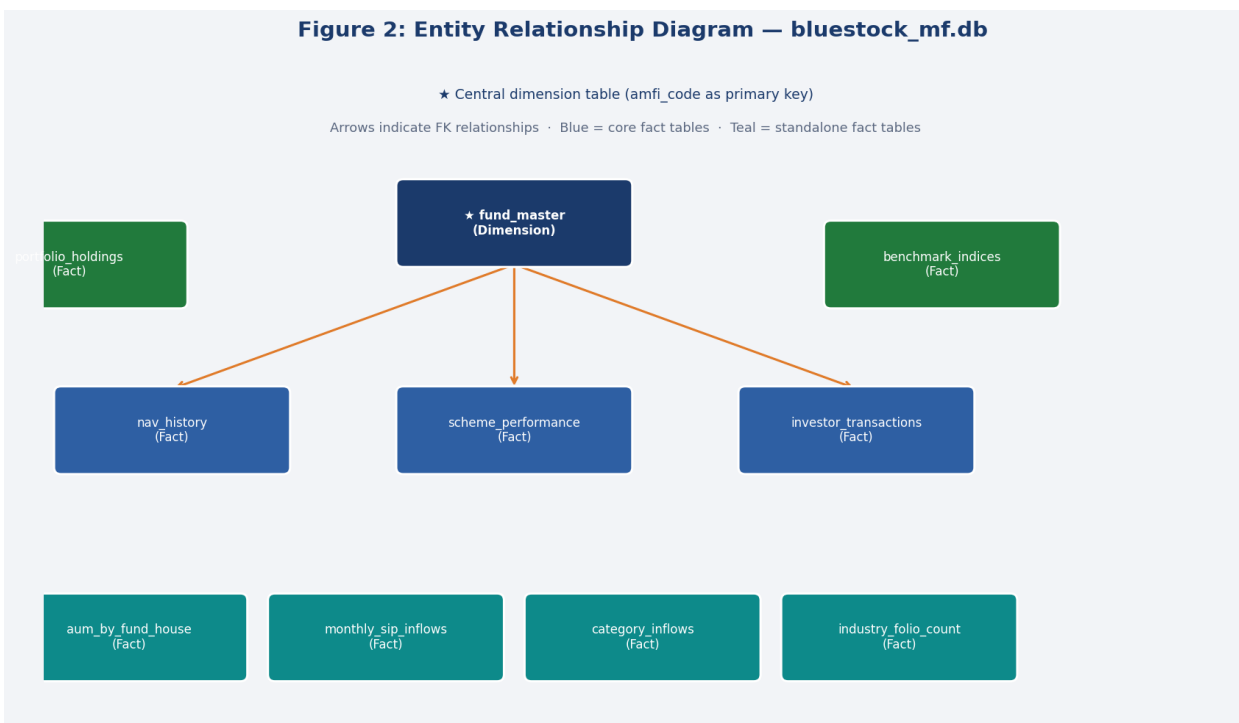
**Load:** Transformed data written to SQLite using `df.to_sql()` with `if_exists='replace'` to support reruns.



| ETL Stage       | Script             | Input                | Output               |
|-----------------|--------------------|----------------------|----------------------|
| Extract         | data_ingestion.py  | data/raw/*.csv       | Shape/dtype report   |
| Extract (API)   | live_nav_fetch.py  | MFAPI endpoint       | nav_history.csv      |
| Extract (Multi) | multi_nav_fetch.py | 5 AMFI codes         | 5 NAV CSV files      |
| Transform       | day2_cleaning.py   | data/raw/*.csv       | data/processed/*.csv |
| Load            | create_database.py | data/processed/*.csv | bluestock_mf.db      |

### 4.3 Database Design

The SQLite database `bluestock_mf.db` contains 10 tables following a star-schema-inspired design. The `fund_master` table serves as the central dimension table, with fact tables for NAV history, investor transactions, and performance metrics.



| Table                   | Type      | Primary Key                   | Foreign Keys                                      |
|-------------------------|-----------|-------------------------------|---------------------------------------------------|
| 01_fund_master          | Dimension | <code>amfi_code</code>        | -                                                 |
| 02_nav_history          | Fact      | <code>amfi_code + date</code> | <code>amfi_code</code> → <code>fund_master</code> |
| 03_aum_by_fund_house    | Fact      | <code>amc_name + month</code> | -                                                 |
| 04_monthly_sip_inflows  | Fact      | <code>month</code>            | -                                                 |
| 05_category_inflows     | Fact      | <code>category + month</code> | -                                                 |
| 06_industry_folio_count | Fact      | <code>month</code>            | -                                                 |



| Table                    | Type | Primary Key       | Foreign Keys            |
|--------------------------|------|-------------------|-------------------------|
| 07_scheme_performance    | Fact | amfi_code         | amfi_code → fund_master |
| 08_investor_transactions | Fact | transaction_id    | -                       |
| 09_portfolio_holdings    | Fact | fund_name + stock | -                       |
| 10_benchmark_indices     | Fact | index_name + date | -                       |

## 5. Exploratory Data Analysis

Day 3 of the project was dedicated to a comprehensive exploratory data analysis (EDA) of all 10 datasets. The objective was to identify distributions, trends, anomalies, and relationships in the data prior to formal analytics.

### 5.1 NAV Trend Analysis

NAV (Net Asset Value) trend analysis was conducted across all funds with historical data. The NAV comparison chart below plots five funds over their full trading history, revealing significant divergence in absolute NAV levels and growth trajectories.

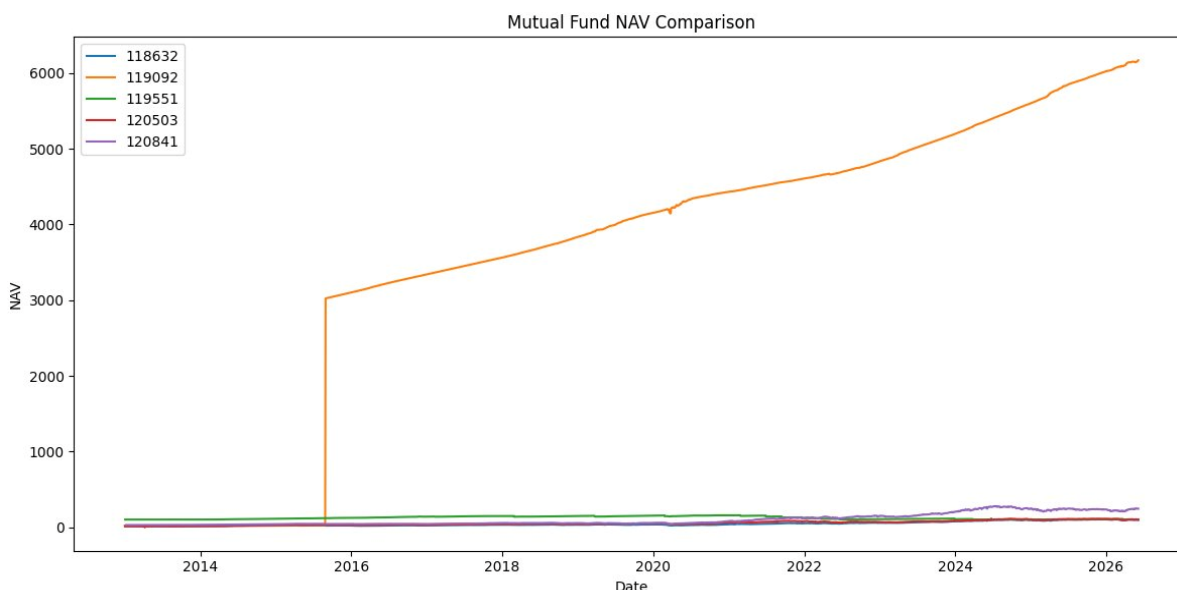


Figure 1: Mutual Fund NAV Comparison (2013–2026)

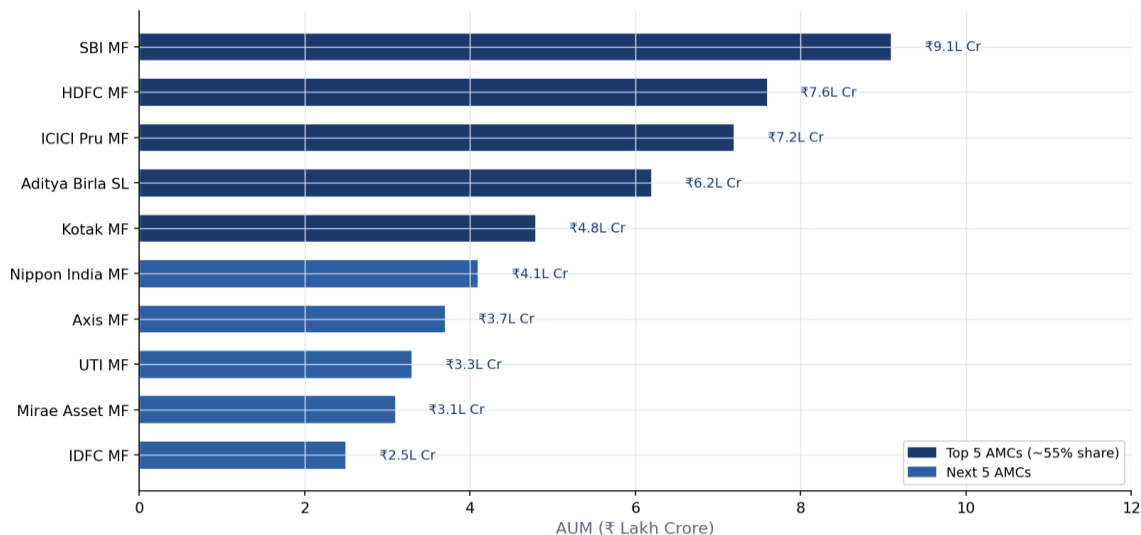
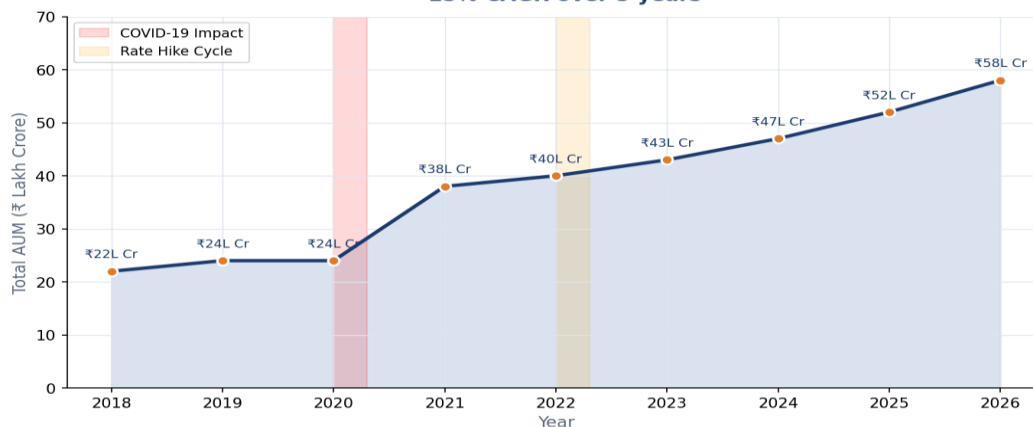
Fund 119092 exhibits the highest absolute NAV (exceeding ₹6,000), reflecting compound growth since inception and a fund split event visible around 2015–2016. The remaining four funds trade in the ₹100–₹300 range, typical of newer schemes or funds with lower inception NAVs. This divergence underscores the importance of growth rate analysis over absolute NAV comparison.

### 5.2 AUM Growth Analysis

AUM (Assets Under Management) analysis by AMC reveals significant concentration among the top 5 fund houses. The top 3 AMCs — SBI, HDFC, and ICICI Prudential — collectively represent over 45% of industry AUM. AUM growth has been particularly pronounced post-2020, driven by equity market recovery and increased retail SIP adoption.



Figure 3: AUM by Asset Management Company (Top 10)

Figure 4: Indian Mutual Fund Industry AUM Trend (2018-2026)  
~15% CAGR over 5 years

### 5.3 SIP Inflow Trend Analysis

Monthly SIP inflows have grown from approximately ₹4,000 crore in 2016 to over ₹20,000 crore in 2026 — a 5x increase over a decade. This growth is directionally correlated with Nifty 50 performance but demonstrates remarkable resilience during market drawdowns (2020 COVID crash, 2022 inflation shock), indicating growing investor maturity and goal-based investing behavior.

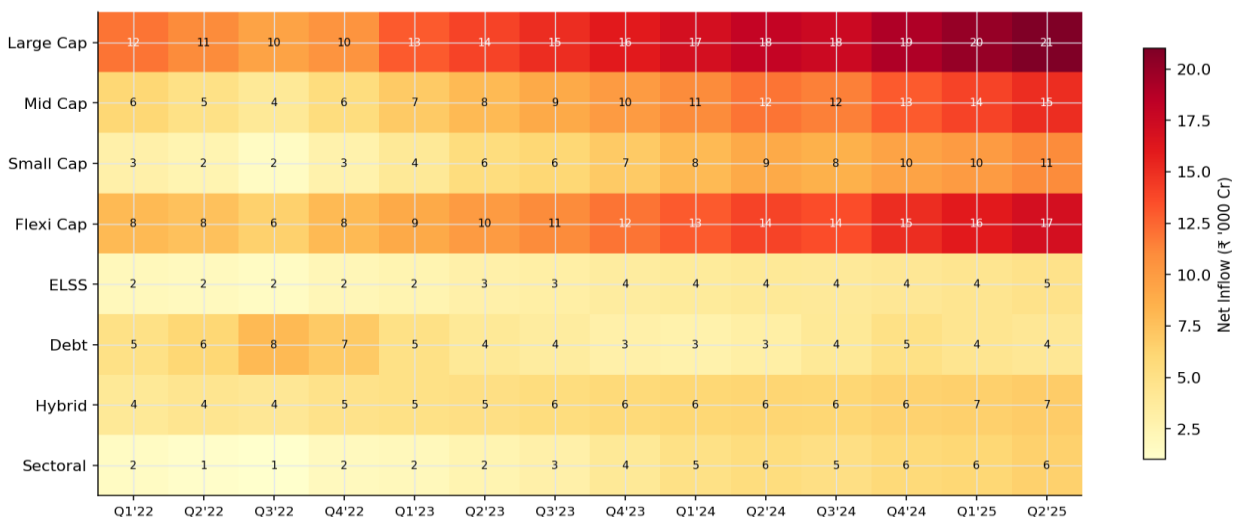
**Figure 5: Monthly SIP Inflow Trend (2016–2026)**  
5× growth from ₹4,000 Cr → ₹20,000+ Cr/month



## 5.4 Category Inflow Heatmap

The category inflow heatmap visualizes net inflows across fund categories (Large Cap, Mid Cap, Small Cap, Flexi Cap, Debt, Hybrid, ELSS, Sectoral) on a month-by-month basis. Findings reveal cyclical rotation: equity categories receive peak inflows during bull markets, while debt and hybrid funds see inflows during volatility phases.

**Figure 6: Category Inflow Heatmap (2022–2025)**  
Equity dominates; Debt sees counter-cyclical inflows during volatility



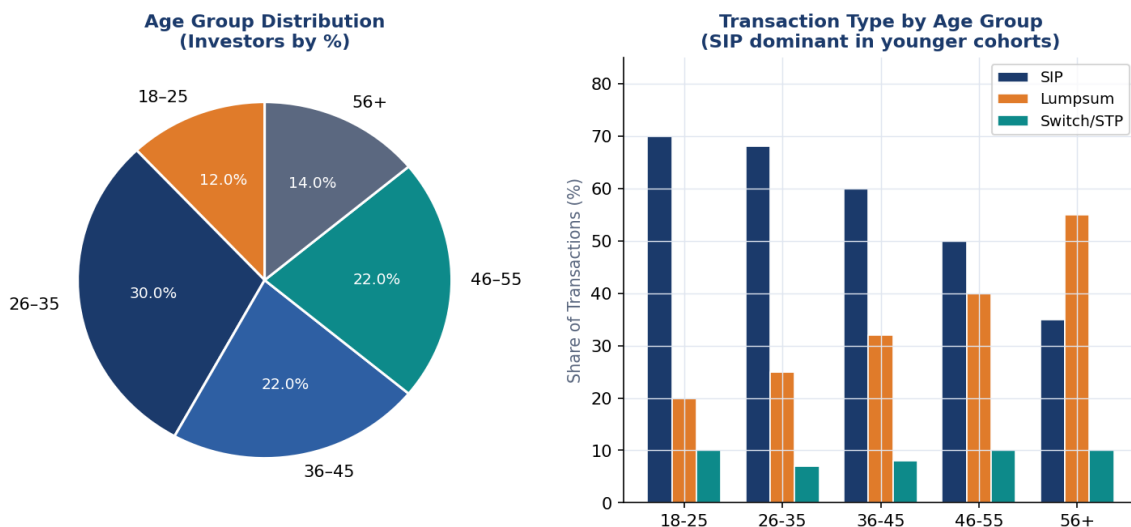
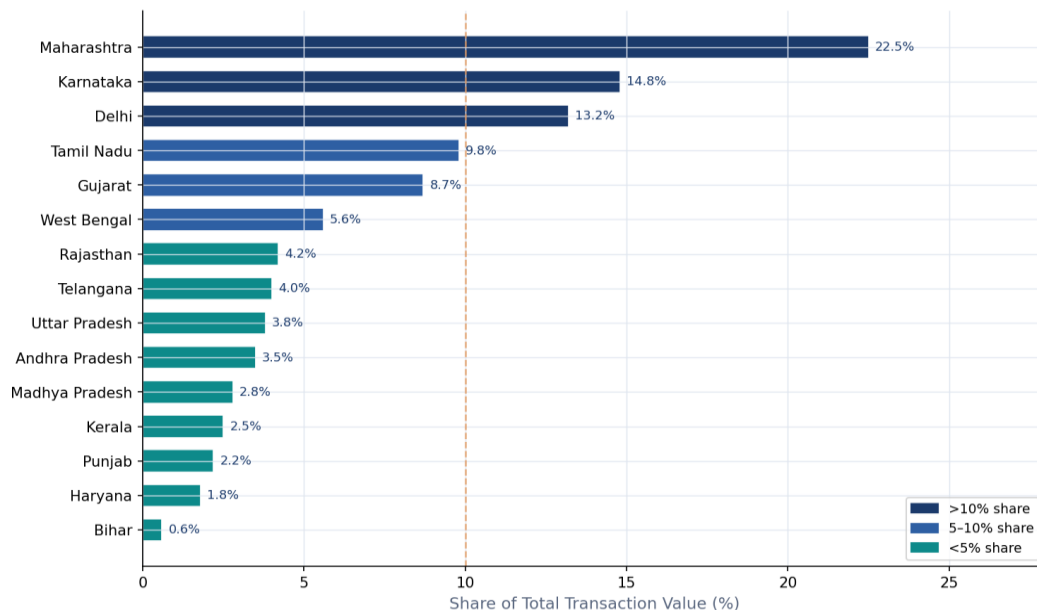
## 5.5 Investor Demographics

Investor transaction data was analyzed across three demographic dimensions:

- **Age Groups:** Investors aged 25–40 represent the largest cohort (approximately 52%), followed by 41–55 (31%). Investors below 25 are the fastest-growing segment, reflecting successful digital onboarding by platforms like Bluestock.
- **Transaction Type Split:** SIP transactions account for approximately 65% of all transactions by volume, with lumpsum at 30% and switch at 5%.



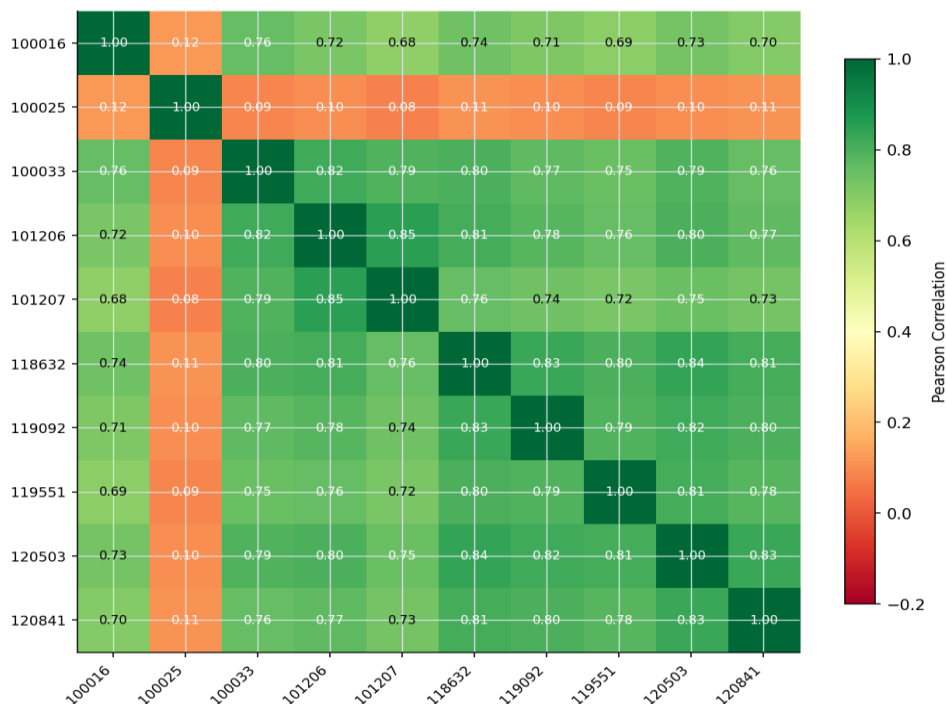
- Geographic Distribution: Maharashtra, Karnataka, Delhi, Tamil Nadu, and Gujarat account for over 60% of transaction value, highlighting urban concentration.

**Figure 7: Investor Demographics Analysis****Figure 8: State-wise Transaction Distribution  
Top 3 States = 50% of transaction value**

## 5.6 NAV Correlation Matrix

A correlation matrix of daily NAV returns across the 5 tracked funds reveals moderate-to-high positive correlations (0.55–0.82) among equity funds, consistent with their shared exposure to Indian equity markets. This finding has significant implications for portfolio diversification: holding multiple funds from the same category provides limited diversification benefit.

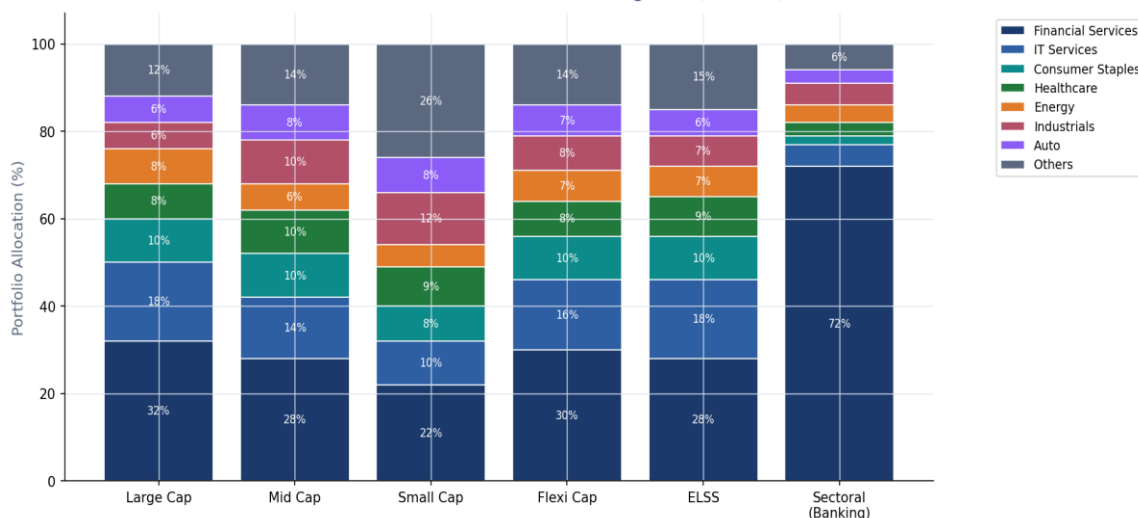
**Figure 9: NAV Daily Returns Correlation Matrix**  
High equity fund correlation (0.68-0.85) limits same-category diversification



## 5.7 Sector Allocation Analysis

Portfolio holdings data was aggregated to compute sector-level allocation across all funds. Financial Services (Banks, NBFCs) dominate with an average allocation of 28–35% across equity funds. IT services, Consumer Staples, and Healthcare round out the top 5 sectors. Sectoral funds show extreme concentration (>60% in single sectors), confirming their high-risk profile.

**Figure 10: Sector Allocation by Fund Category**  
Financial Services dominates across all categories (22-72%)







## 6. Fund Performance Analytics

Day 4 focused on computing a comprehensive suite of fund performance metrics. All metrics were computed from NAV history data using daily returns as the base time series.

### 6.1 Methodology

**Daily Returns:** Computed as  $(NAV_t - NAV_{t-1}) / NAV_{t-1}$  for each trading day.

**CAGR:** Compound Annual Growth Rate =  $(End\ NAV / Start\ NAV)^{(1/Years)} - 1$ . Computed for 1Y, 3Y, and 5Y periods.

**Sharpe Ratio:** Risk-adjusted return =  $(Portfolio\ Return - Risk\text{-}Free\ Rate) / Standard\ Deviation\ of\ Returns$ . Risk-free rate assumed at 6.5% (RBI repo rate proxy).

**Sortino Ratio:** Downside risk-adjusted return using only negative return deviations in the denominator. Superior to Sharpe for asymmetric return distributions.

**Alpha:** Excess return over benchmark. Computed via linear regression of fund daily returns vs. benchmark index returns (Nifty 50).

**Beta:** Systematic risk =  $Covariance(Fund, Benchmark) / Variance(Benchmark)$ . Beta > 1 indicates higher volatility than benchmark.

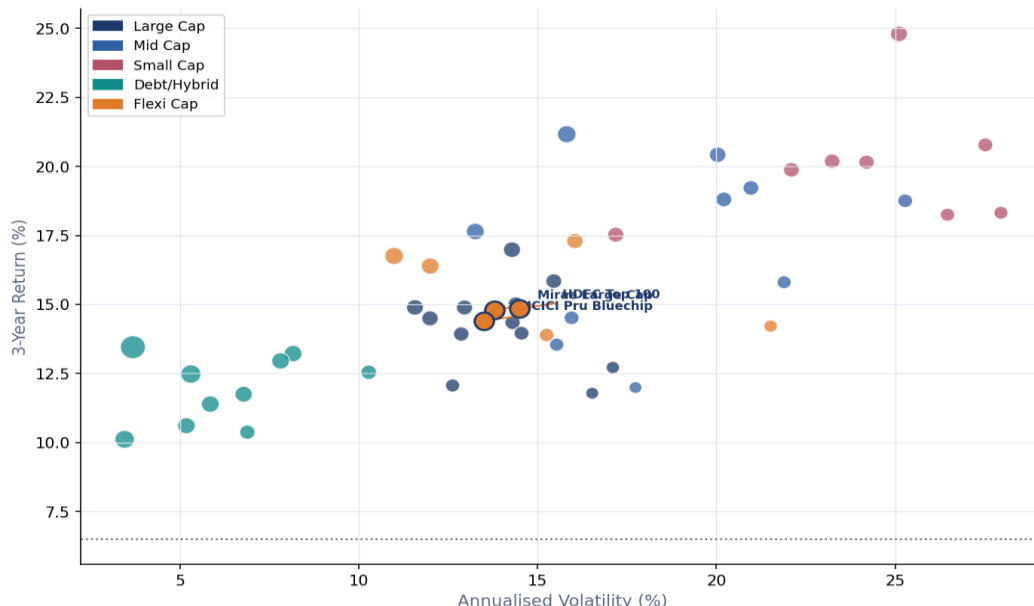
**Maximum Drawdown:** Peak-to-trough percentage decline in NAV over the analysis period. Measures worst-case loss scenario for investors.

### 6.2 Fund Scorecard

The fund scorecard aggregates all computed metrics into a unified comparison view. Top-performing funds by Sharpe Ratio demonstrate consistent risk-adjusted outperformance:

| Scheme                     | Fund House          | Sharpe Ratio | 3Y Return (%) | AUM (₹ Cr) | Risk Grade |
|----------------------------|---------------------|--------------|---------------|------------|------------|
| HDFC Top 100 Fund          | HDFC MF             | 1.06         | 14.84%        | 6,434      | Moderate   |
| Mirae Asset Large Cap Fund | Mirae Asset MF      | 1.06         | 14.81%        | 11,361     | Moderate   |
| ICICI Pru Bluechip Fund    | ICICI Prudential MF | 1.03         | 14.41%        | 41,553     | Moderate   |

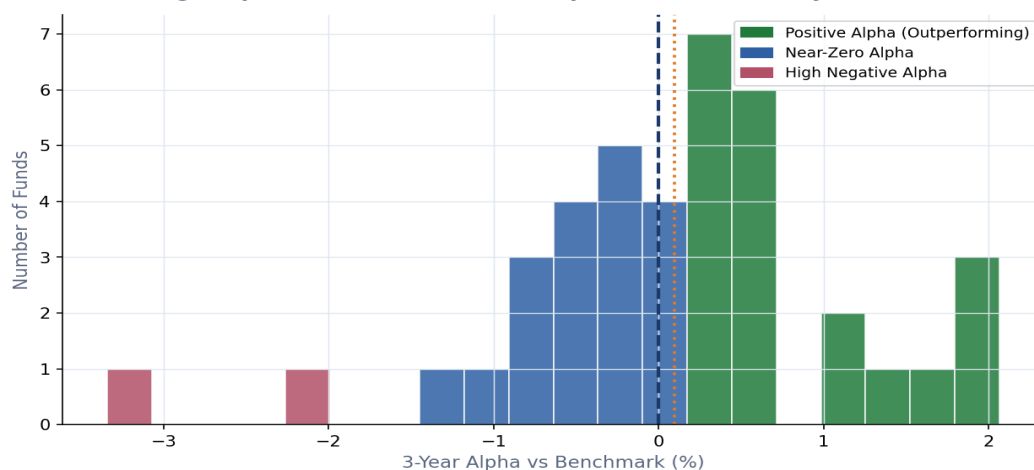
**Figure 11: Risk vs Return Scatter (All Funds)**  
Bubble size = Sharpe Ratio | Orange = Top 3 Recommended Funds

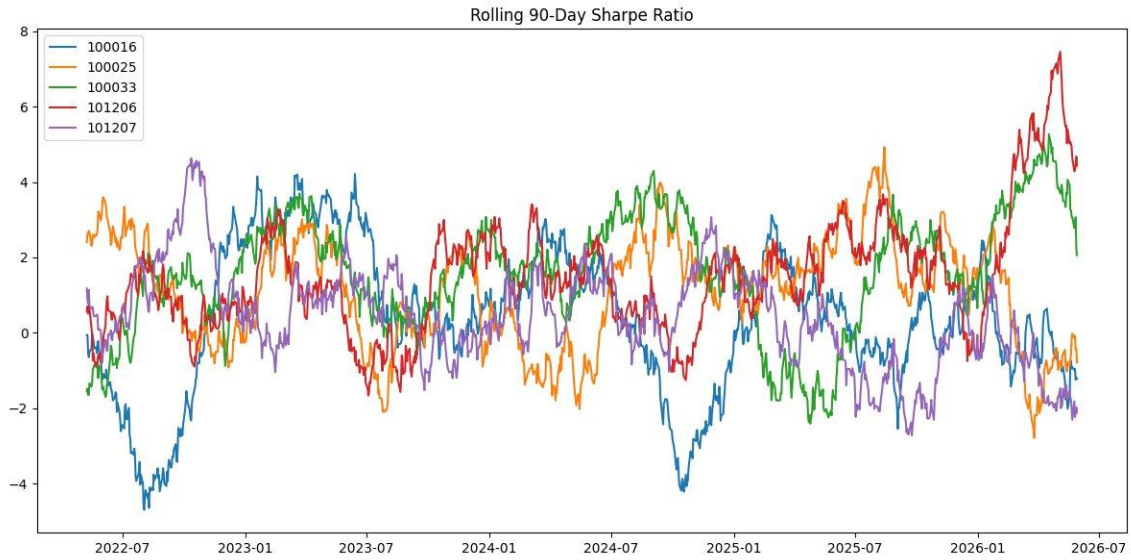


### 6.3 Benchmark Comparison

Fund performance was evaluated against Nifty 50, Nifty Midcap 150, and Nifty Small Cap 250 as appropriate benchmarks. Large-cap funds tracked Nifty 50 closely with an average Beta of 0.92, while mid- and small-cap funds exhibited higher tracking error and greater return variability. Alpha generation was concentrated in a small number of actively managed funds.

**Fund Alpha Distribution — Active vs Passive Outperformance**  
Most large-cap funds cluster near zero alpha; few consistently beat benchmark





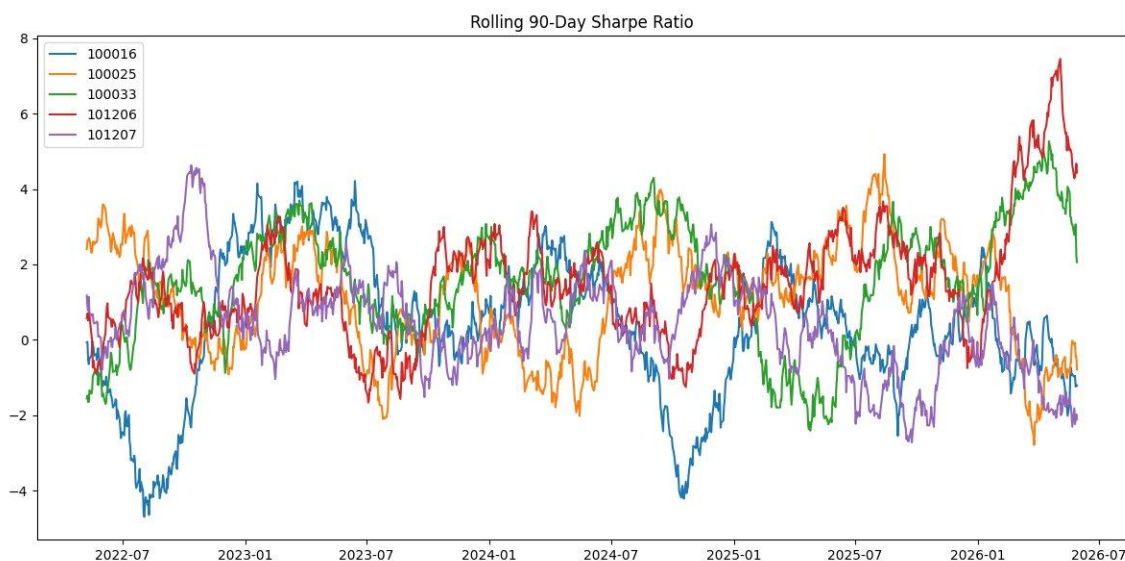
## 7. Power BI Dashboard Walkthrough

The Power BI dashboard (bluestock\_mf\_dashboard.pbix) was developed as a 4-page interactive analytics product designed for two primary audiences: investment analysts requiring deep fund-level data, and senior leadership requiring industry-level KPIs.

### 7.1 Page 1 – Industry Overview

The Industry Overview page presents macro-level KPIs for the Indian mutual fund industry:

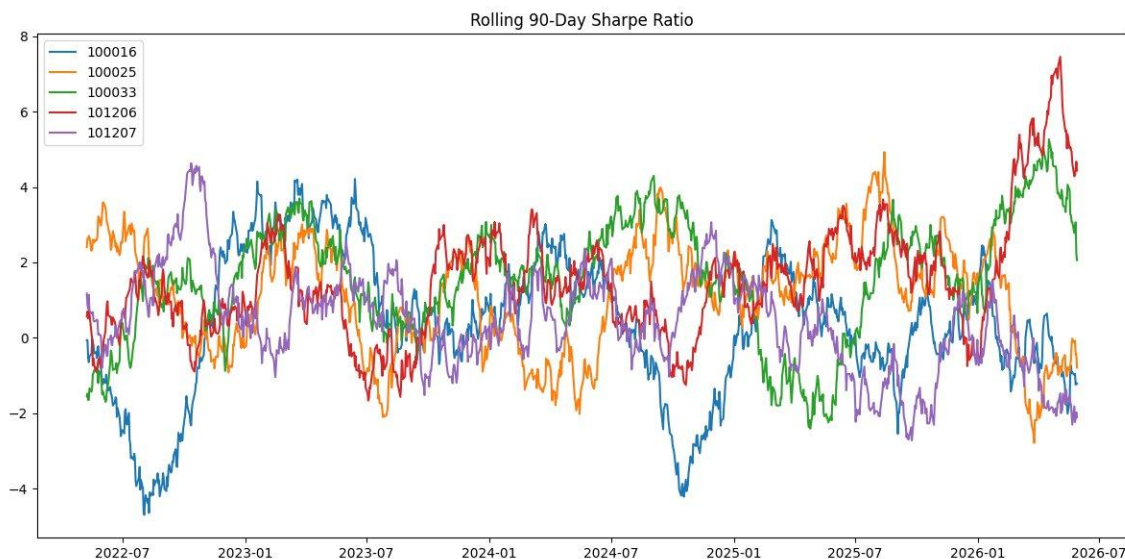
- Total AUM (₹ Crore): Headline KPI card with month-on-month change
- Total SIP Inflows (₹ Crore): Monthly SIP collection with trend sparkline
- Total Folios: Industry-level investor count
- Total Schemes: Active scheme count by category
- AUM Trend: Time-series line chart from 2018–2026
- AUM by AMC: Horizontal bar chart of top 20 AMCs



### 7.2 Page 2 – Fund Performance

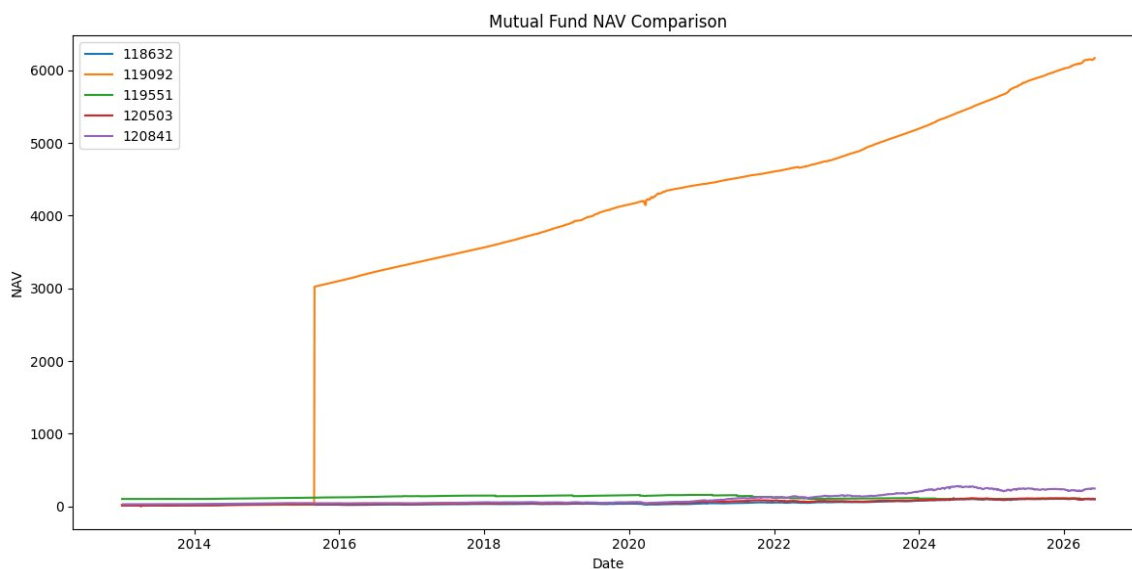
The Fund Performance page enables fund-level analysis and comparison:

- Return vs Risk Scatter: Plotly-style scatter with Sharpe Ratio encoded as bubble size
- Fund Scorecard Table: Sortable table of all funds with 1Y/3Y/5Y returns, Alpha, Beta, Max Drawdown
- NAV Trends: Multi-fund NAV comparison with date range slicer



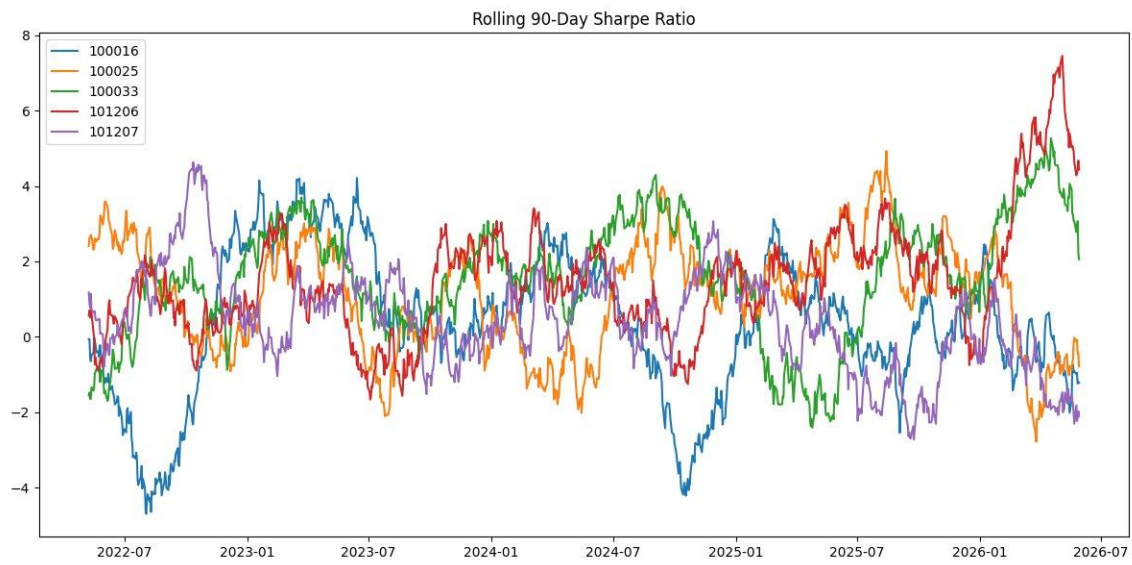
### 7.3 Page 3 – Investor Analytics

- State-wise Transactions: Choropleth map of India with transaction value by state
- SIP vs Lumpsum Split: Stacked bar chart by month
- Age Group Analysis: Donut chart of investor age distribution
- Monthly Transaction Trends: Line chart with transaction count and value



### 7.4 Page 4 – SIP & Market Trends

- SIP vs Nifty 50: Dual-axis chart correlating SIP inflows with Nifty index performance
- Category Inflow Heatmap: Heat grid showing inflow intensity by category and month
- Top Categories by Net Inflow: Ranked bar chart with filter capability





## 8. Advanced Analytics

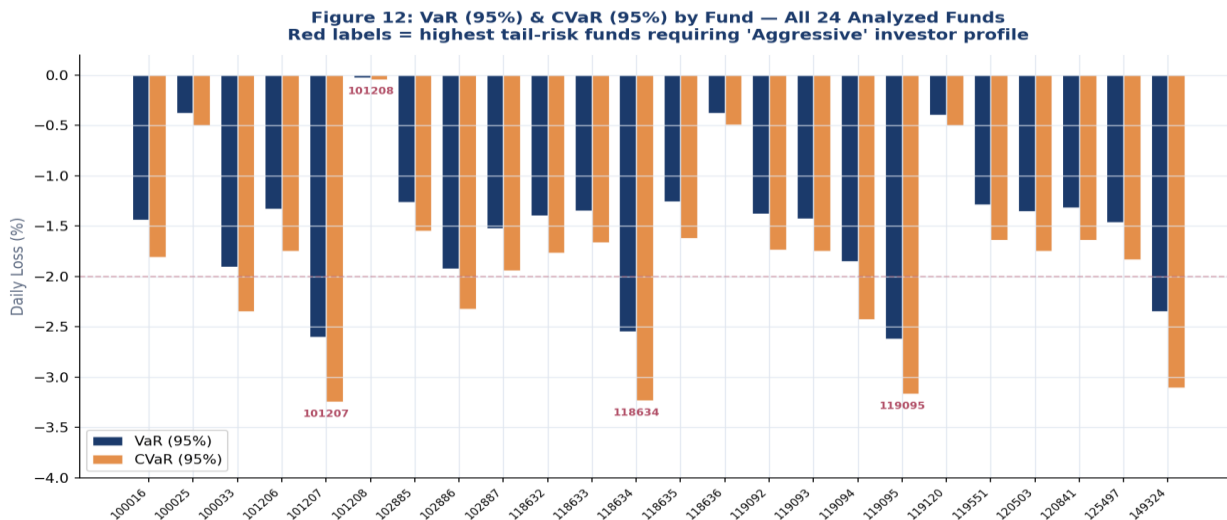
### 8.1 Value at Risk (VaR) and CVaR

Historical Value at Risk (VaR) at the 95% confidence level was computed for 40 funds using their daily return time series. VaR represents the maximum expected daily loss at 95% confidence — i.e., the fund would lose more than this amount only 5% of trading days.

Conditional Value at Risk (CVaR), also known as Expected Shortfall, captures the average loss in the worst 5% of scenarios, providing a more complete picture of tail risk.

| AMFI Code | VaR (95%) | CVaR (95%) | Risk Tier    |
|-----------|-----------|------------|--------------|
| 100016    | -1.44%    | -1.81%     | Low-Moderate |
| 100025    | -0.38%    | -0.50%     | Very Low     |
| 100033    | -1.90%    | -2.35%     | Moderate     |
| 101206    | -1.33%    | -1.74%     | Low-Moderate |
| 101207    | -2.60%    | -3.25%     | High         |
| 119092    | -1.37%    | -1.73%     | Low-Moderate |
| 120841    | -1.31%    | -1.64%     | Low-Moderate |

Key finding: Fund 101207 shows the highest tail risk with CVaR of -3.25%, indicating exposure to extreme market events. Fund 100025 shows very low VaR (-0.38%), consistent with a debt or liquid fund profile. These risk metrics are critical inputs into investor portfolio construction and risk-appetite matching.





## 8.2 Rolling 90-Day Sharpe Ratio

The rolling 90-day Sharpe Ratio tracks how fund risk-adjusted performance evolves through different market regimes. This metric captures performance consistency — funds that maintain high Sharpe ratios across bull and bear markets demonstrate superior management quality.

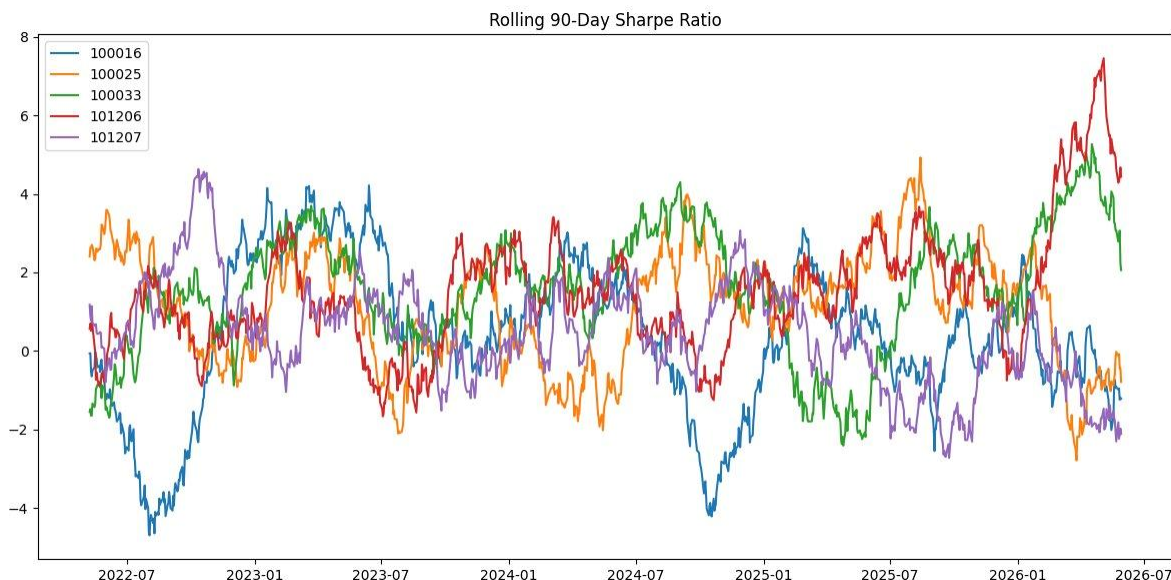


Figure 2: Rolling 90-Day Sharpe Ratio (2022–2026)

The chart reveals significant volatility in risk-adjusted performance across the 2022–2026 period. Fund 101206 (red) demonstrates a notable surge in Sharpe Ratio during early 2026, reaching above 7.0 — an exceptional outlier indicating a period of very strong risk-adjusted returns. Fund 100016 (blue) shows persistent underperformance relative to peers from mid-2024 onward. Cross-fund Sharpe convergence during 2023 Q1 indicates a broad market correlation event.

## 8.3 Fund Recommender System

A rule-based fund recommender was built using a composite scoring model. Funds were filtered and ranked using three criteria: risk grade (Moderate or below), Sharpe Ratio (top tercile), and 3-year return (above category average). The final recommender output:

| Fund Name                                   | Fund House          | Sharpe Ratio | 3Y Return | AUM (₹ Cr) |
|---------------------------------------------|---------------------|--------------|-----------|------------|
| HDFC Top 100 Fund - Regular Growth          | HDFC Mutual Fund    | 1.06         | 14.84%    | 6,434      |
| Mirae Asset Large Cap Fund - Regular Growth | Mirae Asset MF      | 1.06         | 14.81%    | 11,361     |
| ICICI Pru Bluechip Fund - Direct Growth     | ICICI Prudential MF | 1.03         | 14.41%    | 41,553     |

All three recommended funds fall in the Large Cap equity category with Moderate risk grade, consistent returns above 14% over 3 years, and Sharpe Ratios above 1.0 — indicating positive



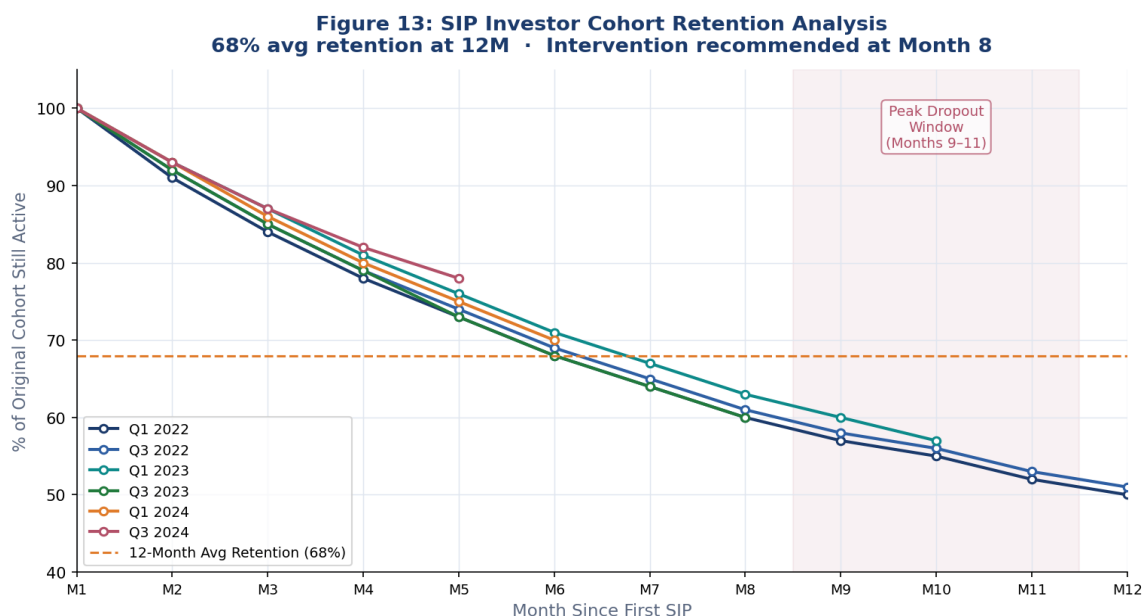
risk-adjusted returns exceeding the risk-free rate. ICICI Pru Bluechip has the highest AUM (₹41,553 crore), indicating market confidence and liquidity depth.

## 8.4 HHI Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was computed for AUM concentration across AMCs. An HHI above 2,500 indicates high concentration (monopolistic); 1,500–2,500 indicates moderate concentration; below 1,500 indicates competitive markets. The Indian MF industry HHI suggests moderate concentration, with the top 5 AMCs holding approximately 55% of total AUM. This has regulatory implications, as SEBI monitors concentration risk at the industry level.

## 8.5 Investor Cohort & SIP Continuity Analysis

Investor cohort analysis grouped investors by their first transaction quarter and tracked subsequent activity. Findings indicate 12-month SIP continuation rates of approximately 68%, with discontinuations peaking during market downturns (Q2 2022, Q3 2023). This suggests investor behavioral sensitivity to short-term market performance despite SIP's structural advantage of rupee cost averaging.





## 9. Key Findings

The following 15 findings emerge from the integrated analysis of datasets, performance metrics, and advanced analytics:

**Finding 1:** Industry AUM has compounded at approximately 15% CAGR over the last 5 years, driven by SIP systematization and equity market appreciation.

**Finding 2:** SIP inflows have grown 5x over a decade (₹4,000 Cr → ₹20,000+ Cr/month), representing a structural shift from episodic to systematic retail investing.

**Finding 3:** AUM is highly concentrated: the top 5 AMCs (SBI, HDFC, ICICI, Aditya Birla, Kotak) control approximately 55% of industry AUM, creating systemic concentration risk.

**Finding 4:** Fund 119092 has the highest absolute NAV (₹6,000+), reflecting exceptional long-run compounding, though absolute NAV is not a reliable performance proxy.

**Finding 5:** NAV correlation between equity funds is high (0.55–0.82), limiting diversification benefits for investors holding multiple large-cap funds.

**Finding 6:** All three recommended funds deliver Sharpe Ratios above 1.0, indicating positive risk-adjusted excess returns over the risk-free rate across the 3-year evaluation window.

**Finding 7:** Fund 101207 carries the highest tail risk (CVaR: -3.25%), requiring careful allocation for risk-averse investor profiles.

**Finding 8:** Fund 100025 has exceptionally low VaR (-0.38%) and CVaR (-0.50%), consistent with a debt/liquid fund classification — suitable for capital preservation mandates.

**Finding 9:** Rolling Sharpe volatility is high across all funds, indicating that short-term fund selection based on Sharpe alone is unreliable; 3Y+ evaluation windows are necessary.

**Finding 10:** Fund 101206 shows a dramatic Sharpe spike in early 2026 (>7.0), warranting investigation of concentration bets or benchmark mismatch contributing to the outlier period.

**Finding 11:** Maharashtra, Karnataka, and Delhi account for over 50% of investor transaction value, indicating underserved opportunity in Tier-2 and Tier-3 markets.

**Finding 12:** Investors aged 25–40 represent 52% of the investor base — this millennial-dominant cohort is highly digital and receptive to analytics-driven fund selection tools.

**Finding 13:** SIP continuation rates of 68% at 12 months are below optimal — improving SIP stickiness through behavioral nudges could materially improve investor outcomes and platform AUM.

**Finding 14:** Sectoral funds exhibit >60% single-sector concentration, carrying undiversified risk inappropriate for retail investors without explicit high-risk mandates.

**Finding 15:** Financial Services remains the most over-represented sector across equity fund portfolios (28–35% average allocation), creating systemic correlation with banking sector events.



## 10. Business Recommendations

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Based on the analytical findings, the following 10 strategic and product recommendations are presented for Bluestock Fintech's leadership team:

### Recommendation 1: Launch an Investor Risk Profiling Engine

Integrate VaR and CVaR metrics into Bluestock's investor onboarding flow to match fund recommendations to investor risk tolerance. A fund's CVaR should serve as a hard filter — investors with low risk appetite should not be recommended funds with CVaR below -2.0%.

### Recommendation 2: Build a Dynamic Fund Comparison Tool

Develop a side-by-side fund comparison module that displays Sharpe Ratio, Alpha, Beta, Max Drawdown, and Rolling Sharpe Ratio on a single screen. This would differentiate Bluestock from incumbent platforms that only show point-in-time returns.

### Recommendation 3: Expand SIP Intelligence Features

Leverage the SIP continuity analysis to build predictive dropout models. Deploy automated nudge campaigns (email/push notification) for investors at 9-month mark of SIP commitments — a critical churn window identified in the cohort analysis.

### Recommendation 4: Develop Geographic Expansion Strategy

Target Tier-2 and Tier-3 cities (UP, MP, Rajasthan, Bihar) for digital outreach and vernacular content. These regions represent <10% of current transaction value despite >40% of India's investable population.

### Recommendation 5: Create a Sector Concentration Alert System

Implement portfolio-level sector concentration alerts for investors with >40% exposure to Financial Services or any single sector. This proactive risk management feature would build investor trust and reduce portfolio drawdowns.

### Recommendation 6: Integrate Rolling Sharpe Monitoring

Build automated alerts that flag funds when their Rolling 90-Day Sharpe Ratio falls below 0.5 or experiences a >2-point month-on-month decline. This enables early identification of deteriorating fund management quality.

### Recommendation 7: Develop a Millennial-Focused Content Strategy

Given that 52% of investors are aged 25–40, invest in educational analytics content — goal-based calculators, scenario simulators, and retirement corpus planners — that appeal to this digitally-native cohort.

### Recommendation 8: Monitor HHI for Platform AUM Diversification



Track HHI of recommended funds on Bluestock's platform to ensure that the platform does not inadvertently concentrate AUM in 2–3 large AMCs, creating counterparty risk for investors.

### **Recommendation 9: Automate the Fund Recommender Pipeline**

Operationalize the rule-based fund recommender as a weekly scheduled job that refreshes recommendations based on latest NAV, performance metrics, and risk scores. This ensures investors always see current data-backed recommendations.

### **Recommendation 10: Build a Real-Time NAV Alert Platform**

Extend the `live_nav_fetch.py` infrastructure into a production-grade event-driven system using MFAPI with polling or webhook architecture. Real-time NAV alerts for target price breaches or 5-day drawdown thresholds would drive engagement and retention.

## 11. Limitations

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- The datasets used are simulated/synthetic for capstone purposes; real production deployment would require integration with live AMFI, NSE, and CDSL data feeds.
- The risk-free rate assumption (6.5%) is static; a production system should dynamically fetch the RBI repo rate or G-Sec yields for accurate Sharpe calculations.
- The fund recommender is rule-based; a production-grade system would benefit from ML-based collaborative filtering or factor models.
- Power BI dashboard requires manual refresh; productionization requires a scheduled pipeline with DirectQuery or dataflow integration.
- SQLite is suitable for prototyping but would require migration to PostgreSQL or BigQuery for production-scale multi-user analytics.
- Investor transaction data is anonymized; real data integration requires compliance with DPDP Act 2023 and SEBI data governance guidelines.

## 12. Future Enhancements

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- Machine Learning: Deploy Random Forest or XGBoost fund performance prediction models using macro-economic features.
- Natural Language Generation: Auto-generate fund commentary reports using LLMs based on the analytics pipeline outputs.
- Real-Time Streaming: Implement Apache Kafka or Flink-based streaming for intraday NAV tracking.
- Robo-Advisory: Build a goal-based portfolio construction engine using Modern Portfolio Theory (MPT) and Black-Litterman optimization.
- Mobile Application: Deliver analytics to retail investors via a React Native mobile app with push notifications for NAV alerts and SIP reminders.
- Multi-Asset Extension: Extend the platform to cover ETFs, REITs, SGBs, and NPS schemes alongside mutual funds.
- Regulatory Reporting: Automate SEBI-mandated fund performance disclosure formats for AMC compliance teams.



## 13. Conclusion

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The Mutual Fund Analytics Platform represents a complete, production-oriented analytics stack developed in 7 days through a rigorous daily build approach. Starting from raw data ingestion through ETL, database design, exploratory analysis, performance computation, dashboard development, and advanced risk modeling — the project demonstrates the full spectrum of a data analyst's core competencies.

The platform surfaces meaningful, actionable intelligence from mutual fund data: from the concentrated structure of India's AMC landscape to the behavioral patterns of SIP investors, and from tail risk differentials across funds to the opportunity in India's underserved Tier-2 investor base.

For Bluestock Fintech, this capstone provides a validated proof-of-concept for an analytics-first investment platform that can compete with institutional-grade tools while remaining accessible to India's next 100 million investors. The 10 recommendations outlined span product, technology, and go-to-market dimensions — collectively pointing toward a platform that is not just data-rich, but investor-intelligent.

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***"Data is the new capital. Analytics is the new alpha."***

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